Applying Distance Relay for Voltage Sag Source Detection

Ashok Kumar Pradhan and Aurobinda Routray

***Abstract*—This paper proposes the application of distance relay information for a voltage sag source detection problem. It is found that the magnitude and angle of impedance before and after the sag event clearly indicate on which side of the power-quality (PQ) monitor, the sag source lies. The required information on impedance can be obtained by the PQ monitor using the estimates of distance relay at that point or applying the algorithm of such relay.**

***Index Terms*—Distance relay, power quality (PQ), voltage sag.**

I. Introduction

VOLTAGE SAG is a common power-quality (PQ) problem which may be detrimental to voltage-sensitive loads. Disturbances, such as faults in a power system, lead to voltage sag of different magnitudes at different nodes in the system. Detection and measurement of voltage sag are essential for possible mitigation and further analysis [1]. Another aspect; voltage sag source location is necessary for PQ troubleshooting, diagnosis, mitigation strategy development, and above all, to identify the responsible agency from the penalty point of view. A direction finder for voltage sag is proposed in [2] using disturbance power and energy computed for the whole period of disturbance. Another method using the slope of a line, obtained by least square fitting approach, is proposed for sag direction detection where voltage and current phasors and power factor are utilized [3].

Today, power agencies are employing dedicated monitors for PQ events. Attempts are also being made to tap protective relay information for PQ analysis [4]. In this paper, we show that it is possible to detect the source of voltage sag using the seen impedance and its angle before and after the sag. The PQ monitor at a sag affected bus can indicate the direction of sag source applying this technique. The corresponding estimates, required by the PQ monitor to compute the impedance and angle, may be obtained from local distance relay or employing its algorithm. With this approach, position of the sag source in an integrated network can be obtained from simultaneous readings of different PQ monitors. The method is tested for various networks including in the presence of a unified power-flow controller (UPFC).

II. Proposed Method

Most of the transmission systems are equipped with distance relays for protection. A digital form of such relay estimates the voltage and current phasors using signal processing algorithms and computes the seen impedance there from to derive the trip decision. The seen impedance of a relay depends on system configuration [radial, interconnected, flexible ac transmission system (FACTS)], fault resistance, system frequency, load condition, etc. For a system as shown in Fig. 1, for a phase-a-to-ground fault, the seen impedance at any bus by phase-a relay will be [5]

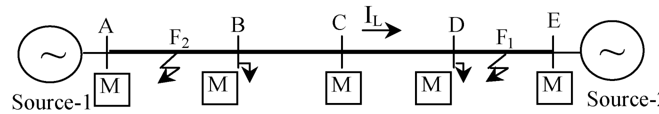

Fig. 1. One-line diagram of the interconnected power system.

$$z_{seen} = \frac{\dot{V}_a}{\dot{I}_a} = z_{1f} + \Delta z \tag{1}$$

where z_{1f} is positive-sequence impedance up to the fault point, and Δz is a function of fault resistance, load angle, etc. For other types of faults, a proper voltage-current pair is to be taken in estimating the impedance.

The above relation refers to the forward direction but in the case of a fault behind the relay, the current direction will be reversed and the resultant seen impedance will change in magnitude and angle both. Therefore, the magnitude and angle of the impedance computed from voltage and current phasors possesses a distinctive feature in identifying the faults/events in front of or behind the relay/monitoring system. The rule for sag source detection becomes

if $|Z_{sag}| < |Z_{presag}|$ and $\angle Z_{sag} > 0 \Rightarrow$ sag source is in front of the PQ monitor, else sag source is behind the monitor. (2)

The difference in the criterion of distance relay and the proposed approach is that the former does not need Z_{presag} for its use. Some of the limitations of distance/directional relay in finding the source of voltage sag are mentioned below.

i) In a radial system and for faults behind the distance relay, there will be no change in seen impedance. As such, no directional relay is also available for such a system.
ii) A transmission line functioning as a part of integrated system may sometimes operate as a radial system fed from either end and in such an event, the information from the directional relay may be wrong for the purpose. This is because at radial system situations, a fault between relay and source does not produce any change for directional relay.
iii) When a fault is not permanent, the local distance relay may not derive a decision.

In this approach, with a fault in the interconnected network, the different PQ monitors use the estimates of the distance relays (or its algorithms) to indicate the direction of the source of sag.

Manuscript received August 21, 2003; revised March 3, 2004. This work was supported by Sponsored Research and Industrial Consultancy, IIT Kharagpur under Innovative Research Scheme-2003. Paper no. PESL-00106-2003.
The authors are with the Department of Electrical Engineering, Indian Institute of Technology, Kharagpur 721302, India (e-mail: akpradhan@ee.iitkgp.ernet.in).
Digital Object Identifier 10.1109/TPWRD.2004.839741

TABLE I
FOR FIG. 1

Sag position	Impedance magnitude(Ω)	Impedance angle (rad)
presag	1897.5	0.001
in front of the monitor	24.1	0.6
behind the monitor	73.4	-2.03

TABLE II
FOR REVERSE POWER FLOW

Sag position	Impedance magnitude(Ω)	Impedance angle (rad)
presag	1889.9	3.13
in front of the monitor	26.6	0.7
behind the monitor	65.8	-2.06

Further, the PQ monitor adjacent to the distance relay, which has derived the trip signal, can also provide the location of the sag source.

III. RESULTS

Faults in front of and behind the monitors are simulated (time domain, with fault resistance $= 10\ \Omega$) at different situations and results are provided with corresponding magnitude and angle of the seen impedance. The impedance is computed from the voltage and current phasors, estimated by Kalman filters that are used for distance relays. Different networks are studied to test the validity of the method and only the results of a few PQ monitors are provided in each case.

A. Case-1 Two-Terminal System

First, we consider the case of the monitor at bus C in Fig. 1. The monitor finds a number of lines on both sides in the system. For the present case, the power flow is from left to right as marked in the figure. Three situations are simulated and corresponding seen impedances are computed i) without fault ($\delta = 0.18$ rad), ii) fault in front of the monitor and beyond the reach of the distance relay at C (F_1), iii) and fault behind the monitor and beyond the reach of the distance relay at C (F_2). The magnitude and angle of the seen impedance of each case as computed by monitor at C is provided in Table I. It is obvious from the results that the impedance angle during sag situation is of positive value for sag source in front of the monitor (0.6 rad) and for behind the monitor case, it is of negative value (-2.03 rad). For both the sag situations, seen impedance is reduced from the presag value.

Another situation of such a system is the reverse power-flow condition (the current direction of I_L in Fig. 1 is changed) and the corresponding results are provided in Table II. These results are also in agreement with (2).

The above results are for a line-to-ground fault. Similar findings are also obtained for all other types of fault. Observations derived by monitors at other affected buses also indicate the direction of fault properly.

B. Case-2 Radial System

The earlier system of Fig. 1 operates now as a radial system as shown in Fig. 2 and its results are presented in Table III. It is observed that for line-to-ground faults behind the monitor at C (F_2), there is no angle or magnitude change in impedance whereas the angle is positive and magnitude of impedance is less than the presag condition for sag source in front of the relay (F_1). These results also satisfy the rule (2).

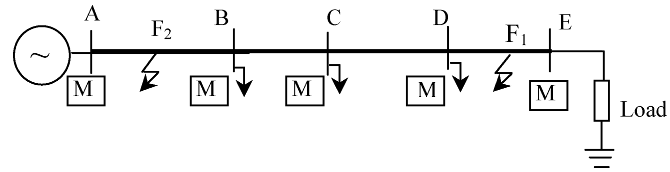

Fig. 2. Radial power system.

TABLE III
RADIAL SYSTEM

Sag position	Impedance magnitude(Ω)	Impedance angle (rad)
presag	101.8	0.47
in front of the monitor (F_1)	12.4	0.73
behind the monitor (F_2)	101.8	0.47

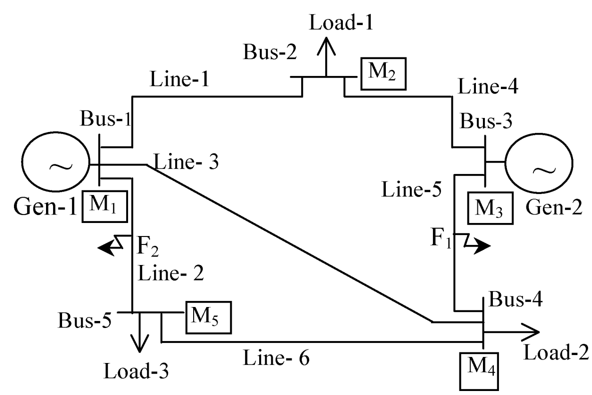

Fig. 3. One-line diagram of the five-bus system.

TABLE IV
FIVE-BUS SYSTEM

Sag position	Impedance magnitude(Ω)	Impedance angle (rad)
presag	925.8	-1.3
in front of the monitor	451.9	1.2
behind the monitor	67.8	-2.16

C. Case-3 Five Bus Power System

A five-bus power system as shown in Fig. 3 is considered next. Two voltage sag situations are observed by the monitor M_5 (at bus 5) by introducing faults at F_1 and F_2 (in front of and behind the monitor, respectively). The results in Table IV depict that the seen impedance angle is a positive value (1.2 rad) for a sag source in front of the monitor and negative value (-2.16 rad) for the sag source behind the monitor. This is in agreement with the logic in (2).

D. Case-4 Transmission System With a UPFC

The sag source locations at F_1 and F_2 (in the double-circuit system of Fig. 4) are in front of monitor M_1 and the angle-results in agreement with the above are positive values of 1.0 and

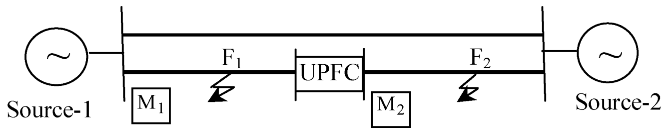

Fig. 4. Double-circuit transmission system with a UPFC.

TABLE V
WITH UPFC

Sag position	M_1 Impedance magnitude(Ω)	M_1 Impedance angle (rad)	M_2 Impedance magnitude(Ω)	M_2 Impedance angle (rad)
presag	436.1	0.87	649.5	0.07
At F_1	12.95	1.0	16.95	-1.0
At F_2	45.97	0.83	28.49	1.1

0.83 rad, respectively. Impedance angles as seen by monitor M_2 for the situations of F_1 and F_2 are -1.0 and 1.1 rad, respectively (see Table V), which indicates that the fault at F_1 is behind the monitor and that at F_2 is in front of it. For each of the sag situations, the impedance magnitude is less than its presag value.

The above results on different systems are only a few of the simulated cases. We studied for different monitoring stations, different types of fault, different situations of systems, variations of fault inception angle, fault resistance, fault duration, etc., and found that the approach based on (2) is accurate in detecting the sag source.

IV. CONCLUSION

A simple approach for sag source detection is proposed in this paper using information from distance relays, protecting lines, or applying their algorithms. It is concluded that obtaining the results of different PQ monitors simultaneously, the sag source can be detected accurately. The test results on various systems indicate its potential to be used in practice.

REFERENCES

[1] J. Arrilaga, M. H. J. Boolen, and N. R. Watson, “Power quality following deregulations,” *Proc. IEEE*, vol. 88, no. 2, pp. 246–261, 2000.

[2] C. Parsons, W. M. Grady, E. J. Powers, and J. C. Soward, “A direction finder for power quality disturbances based upon disturbance power and energy,” *IEEE Trans. Power Del.*, vol. 15, no. 3, pp. 1081–1086, Jul. 2000.

[3] C. Li, T. Tayjasanant, W. Xu, and X. Liu, “Method for voltage-sag-source detection by investigating slope of the system trajectory,” *Proc. Inst. Elect. Eng., Gen. Transm. Distrib.*, vol. 150, no. 3, pp. 367–372, 2003.

[4] D. G. Hart, W. Peterson, D. Uy, J. Schneider, D. Novosel, and R. Wright, “Tapping protective relays for power quality information,” *IEEE Comput. Appl. Power*, vol. 13, no. 1, pp. 45–49, Jan. 2000.

[5] P. K. Dash, A. K. Pradhan, G. Panda, and A. C. Liew, “Adaptive relay setting for Flexible AC Transmission Systems (FACTS),” *IEEE Trans. Power Del.*, vol. 15, no. 1, pp. 38–43, Jan. 2000.